

Ordered pair (Kuratowski, 1921)

$$(x, y) \neq (y, x) \\ \text{if } x \neq y$$

$$(x, y) = \{\{x\}, \{x, y\}\} \\ (y, x) = \{\{y\}, \{y, x\}\}$$

Proposition $(x_1, y_1) = (x_2, y_2) \Leftrightarrow x_1 = x_2 \text{ and } y_1 = y_2$

$$\Rightarrow (x_1, y_1) = (x_2, y_2) \Rightarrow \{\{x_1\}, \{x_1, y_1\}\} = \{\{x_2\}, \{x_2, y_2\}\}$$

$$\textcircled{1} \quad x_1 = y_1 \\ \Downarrow$$

$$\{x_1, y_1\} = \{x_1\} \Rightarrow (x_1, y_1) = \{\{x_1\}, \{x_1\}\} \Rightarrow x_2 = y_2 \\ \Rightarrow x_1 = x_2$$

$$\textcircled{2} \quad x_1 \neq y_1 \text{ and } x_2 \neq y_2 \Rightarrow \{x_1\} \neq \{x_1, y_1\}$$

$$\text{and } \{x_2\} \neq \{x_2, y_2\}$$

$$\{\{x_1\}, \{x_1, y_1\}\} = \{\{x_2\}, \{x_2, y_2\}\} \quad \text{if } \{x_1\} = \{x_2\} \quad \Leftarrow$$

$$\Rightarrow x_1 = x_2 \Rightarrow y_1 = y_2$$

Def The Cartesian product of sets X and Y is the set $X \times Y = \{(x, y) \mid x \in X, y \in Y\}$

$X \times Y = Y \times X$ doesn't hold in general

Def A relation R on the set X is a subset of $X \times X$

Example $X = \{1, 2, 3\}$ $X \times X = \{(1, 1), (1, 2), (1, 3), \dots, (3, 1), (3, 2), (3, 3)\}$

$$R_1 = \{(1, 1), (2, 2), (3, 3)\}$$

$$R_2 = \{(1, 2), (1, 3), (2, 3)\}$$

$$R_3 = \{(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)\}$$

$$R_4 = \{(1, 1)\}$$

$$R_5 = \emptyset$$

$$R_6 = X \times X$$

Def Let $X \neq \emptyset$. A partial order on X is a relation $R \subseteq X \times X$ s.t.

1. R is reflexive, $(x, x) \in R \quad \forall x \in X$

2. R is anti-symmetric.

if $(x, y) \in R$ and $(y, x) \in R \Rightarrow x = y$

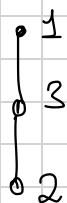
3. R is transitive

if $(x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$

Example $x \in \{1, 2, 3\}$

$$\textcircled{1} \quad 1 \preceq 1, 2 \preceq 2, 3 \preceq 3, 1 \preceq 2, 1 \preceq 3, 2 \preceq 3$$

$$\textcircled{2} \quad (1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (3, 2)$$



Natural numbers

$$\begin{array}{ccccccc} \emptyset & \{\emptyset\} & \{\emptyset, \{\emptyset\}\} & \{\emptyset, \{\emptyset, \{\emptyset\}\}\} & \dots & N_{n+1} = \{N_n\} \cup \{\emptyset\} \\ \parallel & N_1 & N_2 & N_3 & & \\ N_0 & & & & & \end{array}$$

Def Let N be a set with a distinguished element denoted as 0 and a function $s: N \rightarrow N$. We say that $(N, s, 0)$ satisfy Peano axioms if

1. There is no $n \in N$ with $s(n) = 0$
2. The map s is injective: if $s(m) = s(n)$ for some $m, n \in N$, then $m = n$
3. If $K \subseteq N$ is such that $0 \in K$ and if $n \in K \Rightarrow s(n) \in K$, then $K = N$

Lecture 2

09.09

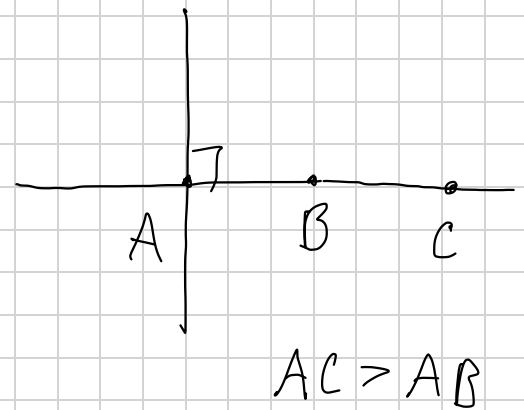
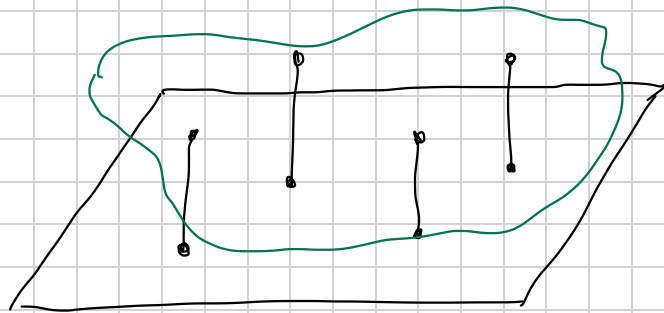
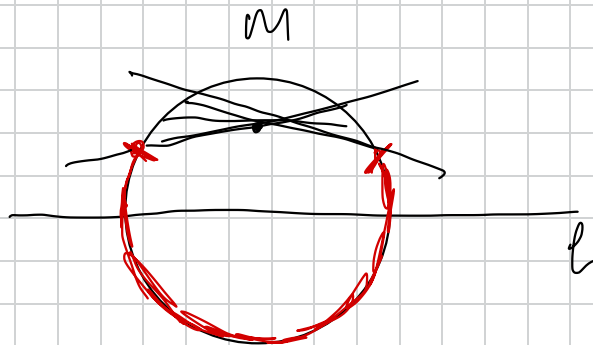
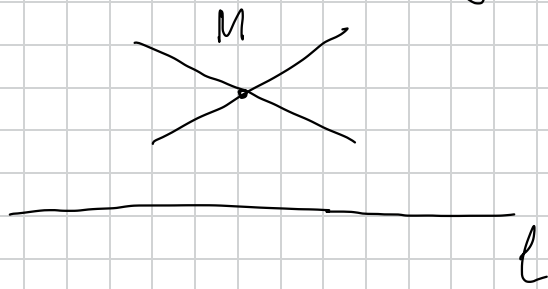
Alternative math

$x \in X$ with probability p

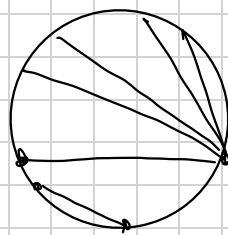
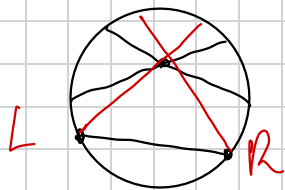
Euclidean geometry

$$\begin{array}{c} M \\ \text{---} \cdot \text{---} \ell' \\ \text{---} \ell \\ \exists ! \ell' : M \in \ell', \ell' \cap \ell = \emptyset \end{array}$$

Hyperbolic geometry



Model:



class of parallel lines

K is inductive if $0 \in K$ and $(n \in K) \Rightarrow (S(n) \in K)$

$K \subseteq \mathbb{N}$. 3rd Peano axiom: if X is inductive, then $K = \mathbb{N}$

Proposition

Every non-zero natural number is a successor

$$\forall n \in \mathbb{N}: n=0 \text{ or } \exists m: S(m)=n$$

Construct $K = \{n \in \mathbb{N} \mid n=0 \text{ or } \exists m \in \mathbb{N}: S(m)=n\}$

$$0 \in K \quad \text{True}$$

assume $n \in K$, prove that $S(n) \in K$

$$S(n) \rightarrow n \in \mathbb{N}, S(n) = S(m), m=n$$

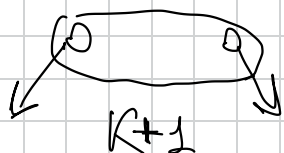
Then K is inductive

$$n = n+1$$

$$n+1 = n+2$$

$$n=1 \quad \text{True}$$

$$n=k \quad \text{True}$$



Arithmetic

Definition

Let X be a set. A binary operation on X is a map

$$\odot: X \times X \rightarrow X$$

$$x \odot y = z$$

$$\odot(x, y) = z$$

Example

$$X = \{a, b\}$$

$$\odot(a, a) = a \quad \odot(a, b) = b \quad \odot(b, a) = b \quad \odot(b, b) = a$$

Addition of natural numbers

Def

For all $m, n \in \mathbb{N}$ define

$$1. \quad m + 0 = m$$

$$2. \quad m + S(n) = S(m + n)$$

$$1 = S(0)$$

$$2 = S(S(0))$$

$$\begin{aligned} 1 + 2 &= 1 + S(1) = S(1 + 1) = S(1 + S(0)) = S(S(1 + 0)) = \\ &= S(S(1)) = S(S(S(0))) = 3 \end{aligned}$$

Proposition

For all $m, n, p \in \mathbb{N}$ the following holds:

1. $m + (n + p) = (m + n) + p$ (associativity)
2. $0 + n = n$ - neutral element
3. If $m + p = n + p$ then $n = m$ - cancellation
4. $m + n = n + m$ - commutativity

Proof

$$1. \quad K = \{ p \in \mathbb{N} \mid m + (n + p) = (m + n) + p, \forall m, n \in \mathbb{N} \}$$

$$0 \in K \quad m + (n + 0) = (m + n) + 0 \rightarrow \text{True}$$

$$p \in K \quad m + (n + p) = (m + n) + p$$

Prove that $m + (n + S(p)) = (m + n) + S(p)$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ m + S(n + p) & & S((m + n) + p) \\ \downarrow & & \parallel \text{ by assumption} \\ S(m + (n + p)) & = & S(m + (n + p)) \end{array}$$



$$4. \quad K = \{ n \in \mathbb{N} \mid \forall m \in \mathbb{N} : m + n = n + m \}$$

$$0 \in K \quad m + 0 = 0 + m \quad \text{True}$$

$$n \in K \quad m + n = n + m$$

$$\begin{aligned} m + S(n) &= S(n) + m \\ \parallel \\ S(m + n) &= S(n + m) \end{aligned}$$

$$K_1 = \{ m \mid \forall n \quad S(n + m) = S(n) + m \}$$

$$0 \in K, \quad S(n+0) = S(n) + 0$$

$$S(n+m) = S(n) + m$$

$$S(n + S(m)) = S(n) + S(m)$$

$$S(S(n+m)) = S(S(n) + m)$$

by assumpt.



Lecture 3

11.09

Def for all $m, n \in \mathbb{N}$ define:

1. $m \cdot 0 = 0$
2. $m \cdot S(n) = m \cdot n + m$

Example

$$m \cdot 1 = m \cdot S(0) = m \cdot 0 + m = m$$

Proposition

Suppose $m, n, p \in \mathbb{N}$. Then

1. $m \cdot (n+p) = (m \cdot n) + (m \cdot p)$ distributivity
2. $m \cdot (n \cdot p) = (m \cdot n) \cdot p$ - associativity
3. Neutral element $1 \cdot m = m \cdot 1 = m$
4. $m \cdot n = n \cdot m$ commutativity

Proof 1. Induction in p

Base: $p=0$

$$m \cdot (n+0) \stackrel{?}{=} (m \cdot n) + (m \cdot 0)$$

$$m \cdot n = m \cdot n \quad \text{True}$$

Step: Assume that $m \cdot (n+p) = (m \cdot n) + (m \cdot p)$

Prove $m \cdot (n+s(p)) = (m \cdot n) + (m \cdot s(p))$

$$\begin{aligned} \text{LHS: } m \cdot (n+s(p)) &= m \cdot s(n+p) = m \cdot (n+p) + m = \\ &= m \cdot n + m \cdot p + m = m \cdot n + m \cdot s(p) \quad \text{True} \end{aligned}$$



Ordering $\mathbb{N}$

Def for $m, n \in \mathbb{N}$ we write $m \leq n$ if $\exists c \in \mathbb{N}$
s.t. $m+c=n$

Proposition

For all $m, n, p \in \mathbb{N}$ we have

1. $m \leq m$ ($c=0$) (reflexivity)
2. If $m \leq n$ and $n \leq m$ then $m=n$ (antisymmetry)
3. If $m \leq n$ and $n \leq p$ then $m \leq p$
4. If $m \leq n$ then $p+m \leq p+n$

Proof

$$2. \quad m \leq n, n \leq m \quad \exists c: m+c=n \quad \& \quad \exists d: n+d=m$$

$$\begin{aligned} m &= \overset{n}{m+c} + d \\ &\Downarrow \\ c+d &= 0 \end{aligned}$$